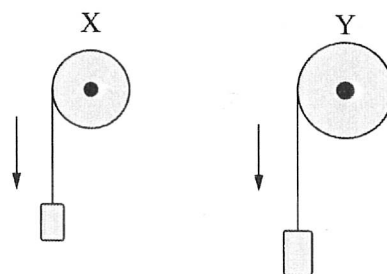


A Side

B

1. Pulleys X and Y are each attached to a block by a string that wraps around the pulley. Both blocks are released from rest and have the same linear acceleration a . As the blocks fall, the pulleys rotate about their centers. Pulley Y has a larger radius than Pulley X. How does the final angular velocity ω_Y of Pulley Y compare to the final angular velocity ω_X of Pulley X?

- (A) $\omega_Y > \omega_X$
 (B) $\omega_Y < \omega_X$
 (C) $\omega_Y = \omega_X$
 (D) It cannot be determined without knowing the relative masses of each block.

A

2. Wheels A and B are connected by a moving belt and are both free to rotate about their centers, as shown. The belt does not slip on the wheels. The radius of Wheel B is three times larger than the radius of Wheel A. Wheel A has constant angular speed ω_A and Wheel B has constant angular speed ω_B . Which of the following correctly relates ω_A

- (A) $\omega_A = \frac{\omega_B}{3}$
 (B) $\omega_A = \omega_B$
 (C) $\omega_A = 3\omega_B$
 (D) $\omega_A = 6\omega_B$

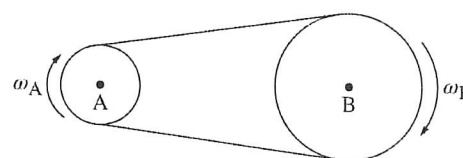


$$V_A = V_B$$

~~$$r_A \omega_A = r_B \omega_B$$~~

$$r_A = \frac{1}{3} r_B$$

$$r_A \omega_A = r_B \omega_B$$

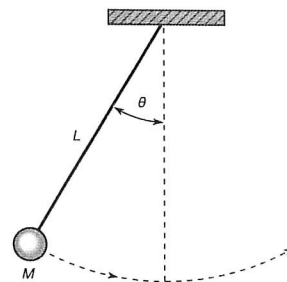
C

3. A pendulum is formed from a mass M connected to a string of length L . The pendulum is initially raised to form an angle θ (measured in radians) and released. The time it takes the pendulum to swing to the opposite side and return to its initial position is measured to be T (measured in seconds). Which of the following is a correct expression for the average linear speed of the pendulum?

- (A) $\frac{\theta}{T}$
 (B) $\frac{\theta L}{T}$
 (C) $\frac{2\theta L}{T}$
 (D) $\frac{4\theta L}{T}$

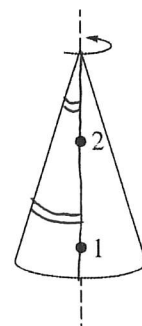
~~$$\frac{\theta L}{T}$$~~

$$\text{arc length } \theta L \times 2$$

D

4. A solid, uniform cone is spinning with constant angular speed about a vertical axis through its center of mass, as shown. A small insect is initially standing on the surface of the cone at Point 1. At a later time, the insect moved to Point 2. Which of the following statements about the motion of the insect is true?

- (A) The angular speed of the insect is greater at Point 1 than at Point 2.
 (B) The angular speed of the insect is greater at Point 2 than at Point 1.
 (C) The linear speed of the insect is greater at Point 1 than at Point 2.
 (D) The linear speed of the insect is greater at Point 2 than at Point 1.



B Side

1. A disk has a radius of 2.5 m and undergoes an angular displacement of $3\pi/4$ radians.

- a. Sketch this angular displacement
b. Convert this measurement to degrees.

$$\frac{3\pi}{4} = \frac{180^\circ \times 3}{4} = 135^\circ$$

- c. Determine the arc length formed.

$$\begin{aligned} \text{arc length} &= \theta \cdot r = \frac{3\pi}{4} \times 2.5 \text{ m} \\ &= \frac{7.5\pi}{4} = \frac{15\pi}{8} = 1.875\pi = 5.89 \text{ m.} \end{aligned}$$

2. A woman walks in a circular path of radius 20 m. The woman undergoes an angular displacement of 2.2 rotations in the counterclockwise direction, which takes her 20 seconds to complete.

- a. Sketch this angular displacement
b. Convert her angular displacement to radians.

$$2.2 \text{ rotation} = 792^\circ = \frac{22\pi}{5}$$

- c. Calculate the woman's linear displacement.

$$360^\circ \times (2.2 - 2) = 72^\circ$$

- d. Calculate the woman's angular velocity (in rad/s).

$$\begin{aligned} \omega &= \frac{v}{r} = \frac{0.11}{20 \text{ m}} = 0.0055 \text{ rad/s} \quad 2.2 \div 20 = 0.11 \text{ rotations/s} \\ \omega &= \frac{\Delta\theta}{t} = \frac{\frac{22\pi}{5}}{20} = 0.22\pi \approx 0.691 \text{ rad/s.} \end{aligned}$$

- e. Calculate the woman's linear velocity (in m/s).

$$\begin{aligned} v &= r\omega = 20 \times 0.22\pi = 4.4\pi \\ &= 13.82 \text{ m/s} \end{aligned}$$

3. A record has a small red dot on its edge. You measure that the dot travels a distance of 1.3 m (clockwise) when the record makes one full rotation, which takes 1.2 seconds.

- a. Determine the radius of the record

$$r = \frac{1.3}{2\pi} = 0.207 \text{ m.}$$

- b. Determine the angular velocity of the record (in deg/s)

$$\omega = \frac{\Delta\theta}{t} = \frac{360^\circ}{1.2} = 300 \text{ deg/s}$$

